

Software Requirements Specification

For

Deepfake Detection System

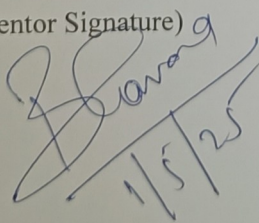
Under the guidance of
Dr. Bhavana Kaushik

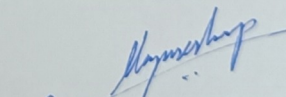
Prepared by

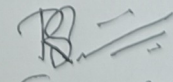
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Introduction

1.1 Purpose of the Project

This document lays out a project plan for the development of Deepfake video detection using neural network. The intended readers of this document are current and future developers working on Deepfake video detection using neural network and the sponsors of the project. The plan will include, but is not restricted to, a summary of the system functionality, the scope of the project from the perspective of the “Deepfake video detection” team (me and my mentors), use case diagram, Data flow diagram, activity diagram, functional and non- functional requirements, project risks and how those risks will be mitigated, the process by which we will develop the project, and metrics and measurements that will be recorded throughout the project.

1.2 Target Beneficiaries

1. Media & Journalism Industry

- News agencies (BBC, CNN, etc.) to verify video authenticity.
- Social media platforms (YouTube, Facebook, Instagram) to detect and flag fake content.

2. Law Enforcement & Cybersecurity

- Police and forensic teams to detect manipulated videos used in crimes.
- Government agencies to prevent misinformation and fraud.

3. Organizations & Corporations

- Companies ensuring brand protection against fake endorsements.
- Financial institutions to prevent fraud in video-based KYC processes.

4. Researchers & Developers

- AI and cybersecurity researchers improving deepfake detection methods.
- Developers working on ethical AI and fake media identification.

5. General Public & Individuals

- People who want to verify the authenticity of videos before believing them.
- Victims of deepfake attacks (fake videos harming reputation).

1.3 Project Scope

1. Development of a Web-Based Platform

- Users can upload videos to classify them as real or fake using a deep learning model.

2. Scalability to Browser Plugins

- The project can be expanded into a browser plugin for automatic deepfake detection while browsing online content.

3. Integration with Social Media Applications

- Platforms like WhatsApp, Facebook, and Instagram can integrate the tool to pre-detect deepfakes before sharing.

4. Enhancements in AI Model & Accuracy

- Continuous improvements in the neural network model to handle new deepfake techniques and increase detection accuracy.

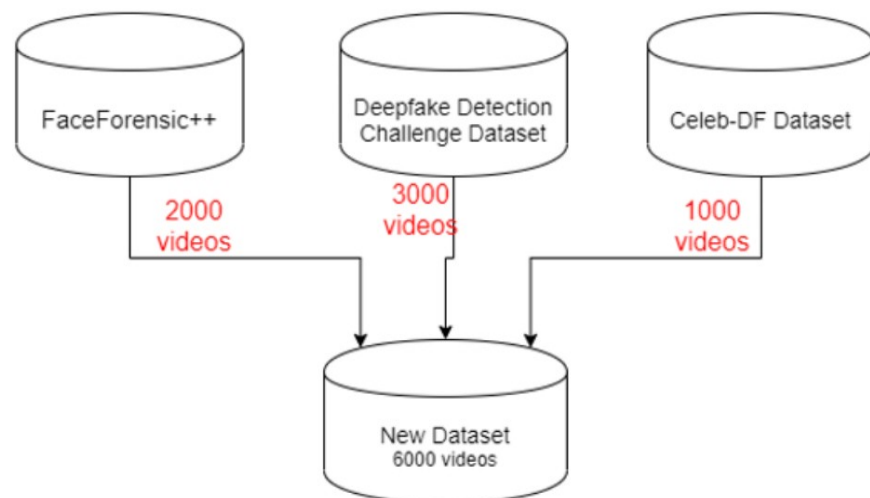
1.4 References

1. **Korshunov, P., & Marcel, S. (2019). Deepfakes: A New Threat to Face Recognition? Assessment and Detection.** *IEEE International Conference on Biometrics Theory, Applications, and Systems (BTAS)*.
2. **Tolosana, R., Vera-Rodriguez, R., Fierrez, J., Morales, A., & Ortega-Garcia, J. (2020). Deepfakes and Beyond: A Survey of Face Manipulation and Fake Detection.** *Information Fusion*, 64, 131-148.
3. **Li, Y., Chang, M. C., & Lyu, S. (2020).** Exposing DeepFake Videos by Detecting Face Warping Artifacts. *IEEE Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*.
4. **Agarwal, S., Farid, H., Gu, Y., He, M., Nagano, K., & Li, H. (2019).** Protecting World Leaders Against Deep Fakes. *IEEE Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*

Project Description

2.1 Dataset

1. We have gathered data from different available datasets like FaceForensic++(FF) [1], Deepfake detection challenge (DFDC) [2], and Celeb-DF [3].
 - a. Further we have mixed the datasets with the collected datasets and created our own new dataset, to accurately and real time detection on different kinds of videos. To avoid the training bias of the model we have considered 50% Real and 50% fake videos.



2.2 Preprocessing Details

- Using glob, we imported all the videos in the directory in a python list.
- cv2.VideoCapture is used to read the videos and get the mean number of frames in each video. GHRCEM-Wagholi, Pune, Department of Computer Engineering 2019-2020 33 Deepfake Video Detection
- To maintain uniformity, based on the meaning a value 150 is selected as an ideal value for creating the new dataset.
- The video is split into frames and the frames are cropped on face location.
- The face cropped frames are again written to new videos using Video Writer.
- The new video is written at 30 frames per second and with the resolution of 112 x 112 pixels in the mp4 format.
- Instead of selecting random videos, to make proper use of LSTM for temporal sequence analysis the first 150 frames are written to the new video

2.3 SWOT Analysis

Strengths

- **High Accuracy:** Utilizes CNNs for precise feature extraction and deepfake detection.
- **Automated & Scalable:** Works on large video datasets with minimal manual intervention.

Weaknesses

- **Computational Complexity:** Requires significant processing power for real-time analysis.
- **False Positives:** Some legitimate videos may be incorrectly flagged.

Opportunities

- **Integration with social media:** Can be used by platforms like YouTube and Facebook for content moderation.
- **Law Enforcement Applications:** Helps forensic teams verify the authenticity of video evidence

Threats

- **Evolving Deepfake Techniques:** Attackers continuously improve deepfake generation models.
- **Ethical Concerns:** Misuse of detection technology could lead to privacy violations.

2.4 Model Details

The model consists of the following layers:

- **ResNext CNN:** The pre-trained model of Residual Convolution Neural Network is used. The model's name is `resnext50_32x4d()`[22]. This model consists of 50 layers and 32 x 4 dimensions.
- **LSTM Layer:** LSTM is used for sequence processing and spot the temporal change between the frames. 2048-dimensional feature vectors are fitted as the input to the LSTM. We are using 1 LSTM layer with 2048 latent dimensions and 2048 hidden layers along with 0.4 chance of dropout, which is capable to do achieve our objective. LSTM is used to process the frames in a sequential manner so that the temporal analysis of the video can be made, by comparing the frame at 't' second with the frame of 't-n' seconds. Where n can be any number of frames before t.

2.5 Hardware Resources Required

In this project, a computer with sufficient processing power is needed. This project requires too much processing power, due to the image and video batch processing.

- **Client-side Requirements:** Browser: Any Compatible Browser Device

Sr. No.	Parameter	Minimum Requirement
1	Intel Xeon E5 2637	3.5 GHz
2	RAM	16 GB
3	Hard Disk	100 GB
4	Graphic card	NVIDIA GeForce GTX Titan (12 GB RAM)

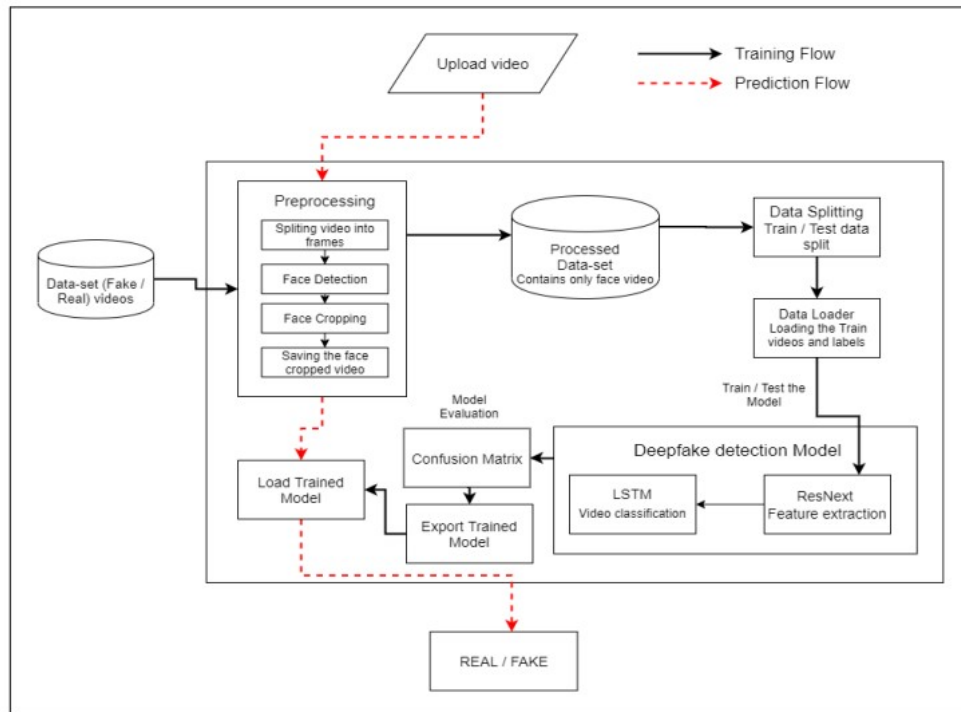
2.6 Software Resources Required

Platform:

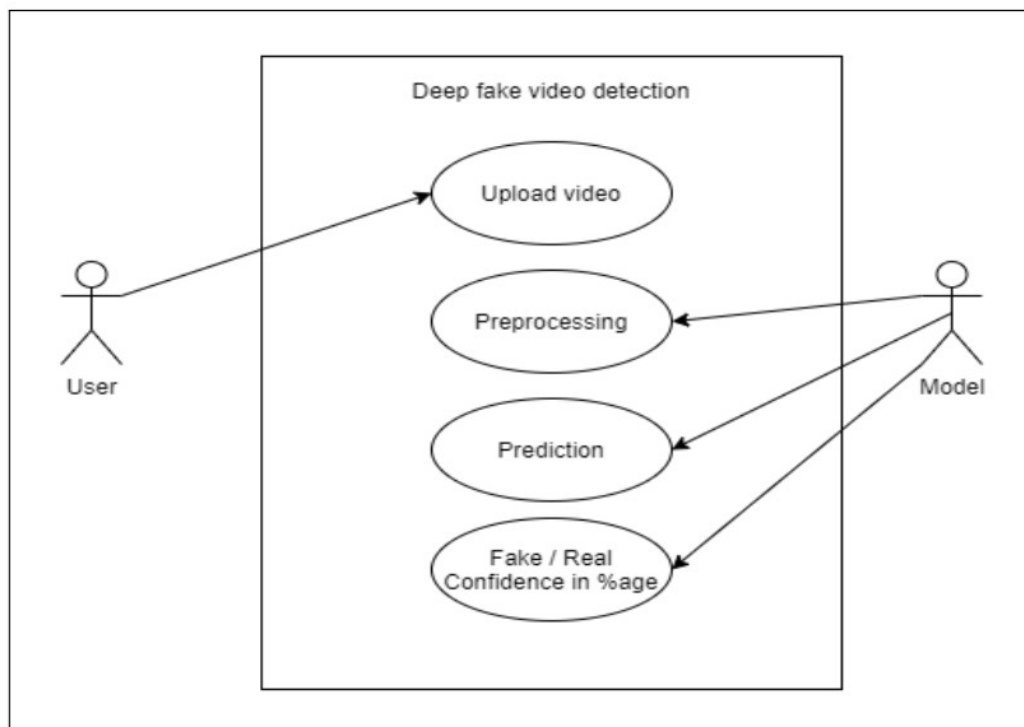
1. Operating System: Windows 7+
2. Programming Language: Python 3.0
3. Framework: PyTorch 1.4
4. Cloud platform: Google Cloud Platform
5. Libraries: OpenCV, Face-recognition

2.7 Design Diagrams

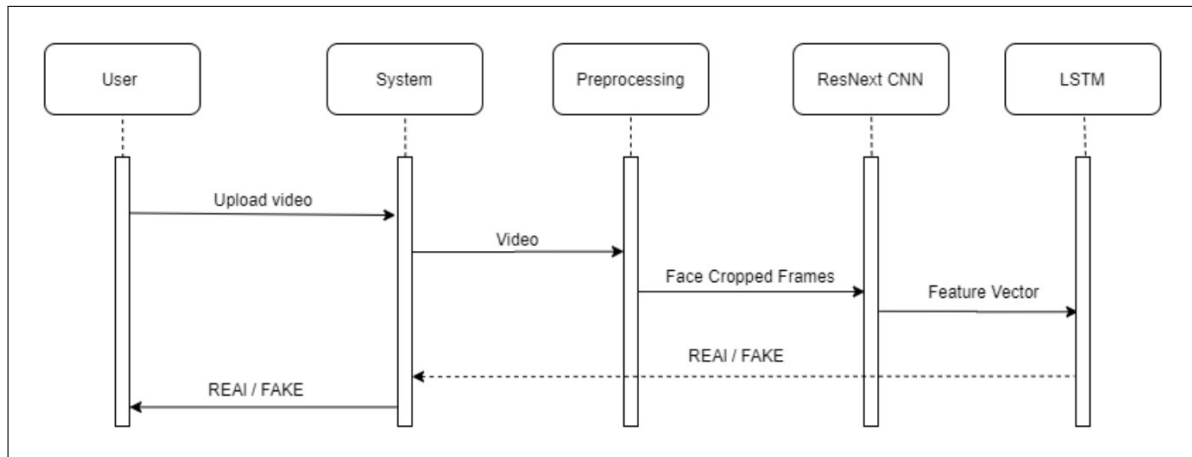
2.7.1 System Architecture



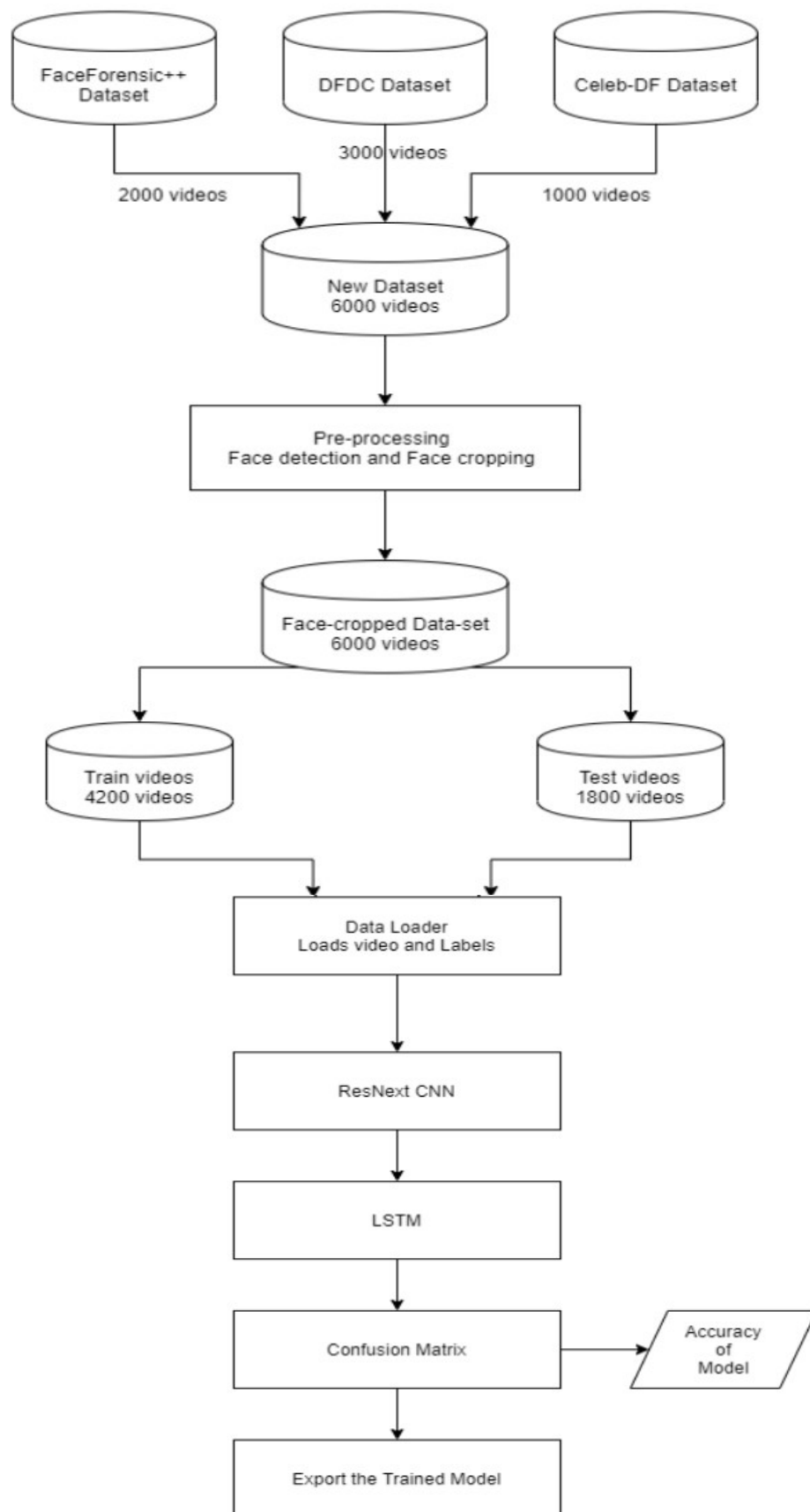
2.7.2 Use Case



2.7.2 Sequence Diagram



2.7.3 Training Workflow:



2.7.3 Pert chart:

Task	Timeframe	Description
Project Initiation(T0)	February 2025	Research and requirement gathering
Dataset Collection & Preprocessing (T1)	February 2025	Gathering and preprocessing deepfake datasets
Model Development (T2)	March 2025	Implementing CNN-based deepfake detection models
Testing & Optimization(T3)	March 2025	Improving model accuracy and efficiency
UI Development (T4)	April 2025	Creating a user-friendly detection interface
Final Testing & Report Submission (T5)	End Semester	Comprehensive evaluation and documentation

Non-functional Requirements

3.1 Performance Requirements

- The software should be efficiently designed to give reliable recognition of fake videos and so that it can be used for a more pragmatic purpose.
- The design is versatile and user friendly.
- The application is fast, reliable and time saving.
- The system has universal adaptations.
- The system is compatible with future upgradation and easy integration.

3.2 Security Requirements

- While uploading the video, the video will be encrypted using a certain symmetric encryption algorithm. On server also the video is in encrypted format only. The video is only decrypted from preprocessing till we get the output. After getting the output the video is again encrypted.
- This cryptography will help in maintain the security and integrity of the video.
- SSL certification is made mandatory for Data security

3.3 Safety Requirement

- The Data integrity is preserved. Once the video is uploaded to the system. It is only processed by the algorithm.
- The videos are kept secured from the human interventions, as the uploaded video is not are able for human manipulation.
- To extent the safety of the videos uploaded by the user will be deleted after 30 min from the server.

Appendix A: Glossary

- **Deepfake:** AI-generated synthetic media that alters or replaces faces in videos to create realistic but fake content.
- **Neural Network:** A machine learning model inspired by the human brain, used in this project to classify videos as real or fake.
- **Dataset:** A collection of videos (both real and fake) used to train and test the deepfake detection model.
- **Preprocessing:** The process of cleaning, resizing, and normalizing video frames before feeding them into the neural network.
- **False Positive:** A real video incorrectly classified as fake.
- **False Negative:** A fake video incorrectly classified as real.

Appendix B: Analysis Model

1. Use Case Diagram for Deepfake Detection:

- Illustrates the interactions between users (video uploaders, system administrators) and the system.
- Shows key functionalities such as uploading videos, processing them, and receiving classification results.

2. Training Workflow Diagram:

- Depicts the end-to-end process of model training, including:
- Data collection (real and fake videos)
- Preprocessing (frame extraction, normalization)
- Model training using neural networks
- Performance evaluation and fine-tuning

3. Sequence Diagram for Video Classification:

- Represents the step-by-step flow of interactions between:
- The user (uploading a video)
- The system (processing frames, running the deepfake detection model)
- The output module (displaying classification results)

4. PERT Chart:

- Provides a structured timeline of project phases, including:
- Dataset collection
- Preprocessing and augmentation
- Model selection and training
- Testing and evaluation
- Deployment and integration

Appendix C: Issues List

1. Dataset Size & Storage:

- **Issue:** Large datasets may cause storage and processing constraints.
- **Status:** Open; exploring cloud-based storage solutions.

2. Accuracy of Deepfake Detection:

- **Issue:** The model's effectiveness depends on dataset diversity and quality.
- **Status:** Ongoing; continuously updating datasets to improve accuracy.

3. User Interface Usability:

- **Issue:** The current interface is minimalistic. A more intuitive GUI is needed for better user experience.
- **Status:** Open; a GUI-based web platform is planned.

4. Scalability:

- **Issue:** The current system may not handle a large volume of video uploads.
- **Status:** Open; optimizing the backend for scalability.

5. Error Handling & False Classifications:

- **Issue:** The model may produce incorrect classifications under certain conditions.
- **Status:** Ongoing; improving model robustness through better training data.

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